

# A Proof-Driven Guide to $q$ -Series, Basic Hypergeometric Identities, Bailey Chains, and Rogers–Ramanujan Theory

A unified article based on the supplied  $q$ -series archive and classical primary literature

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## Abstract

The supplied archive contains a broad collection of notes and formula sheets on  $q$ -Pochhammer symbols, Gaussian coefficients, basic hypergeometric series, Rogers–Ramanujan type identities, Bailey pairs, theta and eta products, partition generating functions, modulus-specific identities, and Borwein-type sign phenomena. This article reorganizes that material into one proof-driven development. Every result labelled as a theorem, proposition, lemma, corollary, or identity is proved here; statements not proved are explicitly labelled as conjectures or research problems.

The main line begins with the finite and infinite  $q$ -binomial theorems and continues through Heine’s transformation, the  $q$ -Gauss and terminating  $q$ -Vandermonde sums, a finite Rogers product lemma, and the  $q$ -Pfaff–Saalschütz summation. These results are then used to prove Bailey inversion, Bailey’s lemma, its limiting forms, and a parameter-lowering step. From this machinery we obtain complete analytic proofs of both Rogers–Ramanujan identities and the entire Andrews–Gordon family. A separate argument derives the Rogers–Ramanujan continued fraction from finite continuants. Jacobi’s triple product, the quintuple product, Poisson summation, theta transformations, the Dedekind eta transformation, and sharp  $q \rightarrow 1^-$  asymptotics are proved as well. The final sections explain how the moduli 7, 9, 14, and 27 material in the archive fits into the general theory, and formulate carefully separated open directions involving finite analogues, Bailey lattices, parity restrictions, and coefficient-sign problems.

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## 1 How the source material is organized here

The archive mixes several conventions: some files use  $q$  for the nome  $e^{\pi i \tau}$ , others use  $q = e^{2\pi i \tau}$ ; some suppress the base in  $(a; q)_n$ ; and the normalization of  ${}_r\phi_s$  is not uniform. We use one convention throughout and translate all identities into it before proving them. The resulting dependency graph is

finite  $q$ -binomial theorem  $\implies$  infinite  $q$ -binomial theorem  $\implies$  Heine and  $q$ -Gauss,  
 finite Cauchy/Rogers identities  $\implies$   $q$ -Pfaff–Saalschütz  $\implies$  Bailey lemma,  
 Bailey chains and a parameter shift  $\implies$  Rogers–Ramanujan and Andrews–Gordon,  
 Euler expansions and Laurent functional equations  $\implies$  triple/quintuple products,  
 Poisson summation plus products  $\implies$  theta/eta transformations and asymptotics.

The modulus-specific files are treated as specializations or extensions of this graph instead of as disconnected lists of formulas.

### Logical and analytic convention

Most finite identities are polynomial or rational identities and therefore hold formally whenever no displayed denominator vanishes. Infinite products and series are first proved for  $|q| < 1$  in their natural domains of absolute convergence. Formal-power-series interpretations are stated separately. When a parameter value is singular in an intermediate formula but the final expression has a removable singularity, the value is understood by a limit.

## 2 $q$ -shifted factorials and Gaussian coefficients

**Definition 2.1** ( $q$ -shifted factorial). For  $n \in \mathbb{N}_0$  and a base  $q$ , put

$$(a; q)_0 = 1, \quad (a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j),$$

and, when  $|q| < 1$ ,

$$(a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

For a list of parameters,

$$(a_1, \dots, a_r; q)_n = \prod_{j=1}^r (a_j; q)_n.$$

The following elementary rules will be used without repeated comment:

$$(a; q)_{m+n} = (a; q)_m (aq^m; q)_n, \tag{2.1}$$

$$(aq^m; q)_n = \frac{(a; q)_{m+n}}{(a; q)_m}, \tag{2.2}$$

$$(a; q)_n = (-a)^n q^{\binom{n}{2}} (q^{1-n}/a; q)_n. \tag{2.3}$$

All three follow directly by rearranging their finite products.

**Lemma 2.2** (Convergence of the infinite product). *Let  $|q| < 1$ . The product  $(a; q)_\infty$  converges locally uniformly as a function of  $a$ , hence defines an entire function of  $a$ . It vanishes exactly at  $a = q^{-m}$ ,  $m \in \mathbb{N}_0$ , with simple zeros when  $q \neq 0$ .*

*Proof.* On a compact set  $|a| \leq M$ ,  $\sum_{j \geq 0} |aq^j| \leq M/(1 - |q|)$ . The standard convergence criterion for products  $\prod(1 + u_j)$ , applied after discarding finitely many factors if necessary, gives uniform convergence. Uniform limits of polynomials are holomorphic. A factor vanishes precisely when  $a = q^{-m}$ ; all remaining factors are nonzero, and the vanishing factor is linear in  $a$ .  $\square$

**Definition 2.3** (Gaussian coefficient). For  $0 \leq k \leq n$  define

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}},$$

and set it equal to 0 outside this range.

**Proposition 2.4** (Pascal recurrences). For  $n \geq 0$  and all integers  $k$ ,

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q, \quad (2.4)$$

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = q^k \begin{bmatrix} n \\ k \end{bmatrix}_q + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q. \quad (2.5)$$

*Proof.* After putting the two terms over the common denominator  $(q; q)_k (q; q)_{n+1-k}$ , the numerator in (2.4) is

$$(q; q)_n \{1 - q^{n+1-k} + q^{n+1-k}(1 - q^k)\} = (q; q)_n (1 - q^{n+1}).$$

This is  $(q; q)_{n+1}$ . The second recurrence follows in the same way, or from the first after the symmetry  $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q$ .  $\square$

**Remark 2.5** (Combinatorial meaning). When  $q$  is a prime power,  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  counts the  $k$ -dimensional subspaces of  $\mathbb{F}_q^n$ . As a polynomial in an indeterminate  $q$ , it is also the rank-generating function of partitions whose Ferrers diagrams fit inside a  $k \times (n - k)$  rectangle. Neither interpretation is needed for the analytic proofs below, but both explain why positivity and convolution identities are natural.

### 3 The finite and infinite $q$ -binomial theorems

**Theorem 3.1** (Finite  $q$ -binomial theorem). For every  $n \in \mathbb{N}_0$ ,

$$(z; q)_n = \sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \quad (3.1)$$

*Proof.* The assertion is clear for  $n = 0$ . If it holds for  $n$ , multiply by  $1 - zq^n$ . The coefficient of  $(-1)^k q^{\binom{k}{2}} z^k$  in the resulting expression is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n+1-k} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q$$

by (2.4). This proves the induction step.  $\square$

**Corollary 3.2** (Finite subset sums). For  $0 \leq k \leq n$ ,

$$\sum_{0 \leq j_1 < \dots < j_k < n} q^{j_1 + \dots + j_k} = q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

*Proof.* Compare the coefficient of  $(-z)^k$  in the product  $\prod_{j=0}^{n-1} (1 - zq^j)$  and in (3.1).  $\square$

**Theorem 3.3** (Infinite  $q$ -binomial theorem). *If  $|q| < 1$  and  $|z| < 1$ , then*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}. \quad (3.2)$$

*The identity also holds coefficientwise in  $\mathbb{C}[a, q][[z]]$ .*

*Proof.* Let  $F(z) = \sum_{n \geq 0} c_n z^n$ , where  $c_n = (a; q)_n / (q; q)_n$ . The ratio

$$\frac{c_n}{c_{n-1}} = \frac{1 - aq^{n-1}}{1 - q^n}$$

is equivalent, coefficient by coefficient, to

$$(1 - z)F(z) = (1 - az)F(qz), \quad F(0) = 1. \quad (3.3)$$

The product  $P(z) = (az; q)_{\infty} / (z; q)_{\infty}$  satisfies the same equation, because

$$\frac{P(z)}{P(qz)} = \frac{1 - az}{1 - z},$$

and  $P(0) = 1$ . Equation (3.3) recursively determines every Taylor coefficient, so  $F = P$  formally. For  $|z| < 1$ , the series converges absolutely by the ratio test and the products converge locally uniformly by Lemma 2.2; hence the formal identity is also analytic.  $\square$

**Corollary 3.4** (Euler's two expansions). *For  $|q| < 1$ ,*

$$\frac{1}{(z; q)_{\infty}} = \sum_{n=0}^{\infty} \frac{z^n}{(q; q)_n}, \quad |z| < 1, \quad (3.4)$$

$$(z; q)_{\infty} = \sum_{n=0}^{\infty} \frac{(-1)^n q^{\binom{n}{2}}}{(q; q)_n} z^n. \quad (3.5)$$

*Proof.* Set  $a = 0$  in (3.2) to obtain (3.4). In (3.2), replace  $z$  by  $z/a$  and let  $|a| \rightarrow \infty$ . For each fixed  $n$ ,

$$(a; q)_n (z/a)^n \longrightarrow (-1)^n q^{\binom{n}{2}} z^n,$$

while the product side tends to  $(z; q)_{\infty}$ . On compact  $z$ -sets the series are dominated by a convergent geometric-majorant refinement, so the limit may be taken termwise. The same argument is valid coefficientwise.  $\square$

**Corollary 3.5** (Reciprocal finite product). *For an integer  $m \geq 0$  and  $|z| < 1$ ,*

$$\frac{1}{(z; q)_{m+1}} = \sum_{n=0}^{\infty} \left[ \begin{matrix} m+n \\ n \end{matrix} \right]_q z^n.$$

*Proof.* Apply (3.2) with  $a = q^{m+1}$ . The product quotient is  $1/(z; q)_{m+1}$ , and

$$\frac{(q^{m+1}; q)_n}{(q; q)_n} = \frac{(q; q)_{m+n}}{(q; q)_m (q; q)_n} = \left[ \begin{matrix} m+n \\ n \end{matrix} \right]_q.$$

$\square$

**Lemma 3.6** (Finite Cauchy augmentation). *For  $n \geq 0$ ,*

$$(xy; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (x; q)_k y^k (y; q)_{n-k}. \quad (3.6)$$

*Proof.* For  $n = 0$  this is immediate. Denote the right-hand side by  $C_n(x, y)$ . Use (2.4), split the sum, shift  $k \mapsto k + 1$  in the second part, and use  $(x; q)_{k+1} = (x; q)_k(1 - xq^k)$  and  $(y; q)_{n+1-k} = (y; q)_{n-k}(1 - yq^{n-k})$ . The two parts combine to

$$C_{n+1}(x, y) = (1 - xyq^n)C_n(x, y).$$

Since  $C_0 = 1$ , induction gives  $C_n = (xy; q)_n$ .  $\square$

## 4 Basic hypergeometric series and Heine's transformation

**Definition 4.1** (Basic hypergeometric series). For parameters away from poles, define

$${}_r\phi_s \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right) = \sum_{n=0}^{\infty} \frac{(a_1, \dots, a_r; q)_n}{(q, b_1, \dots, b_s; q)_n} ((-1)^n q^{\binom{n}{2}})^{1+s-r} z^n.$$

When one upper parameter is  $q^{-N}$ , the series terminates at  $n = N$ .

**Theorem 4.2** (Heine's first transformation). *Assume  $|q| < 1$ ,  $|z| < 1$ , and initially  $|b| < 1$ . Then*

$${}_2\phi_1 \left( \begin{matrix} a, b \\ c \end{matrix}; q, z \right) = \frac{(b, az; q)_{\infty}}{(c, z; q)_{\infty}} {}_2\phi_1 \left( \begin{matrix} c/b, z \\ az \end{matrix}; q, b \right). \quad (4.1)$$

*Both sides extend meromorphically to all parameters for which the displayed expressions are defined.*

*Proof.* Use

$$\frac{(b; q)_n}{(c; q)_n} = \frac{(b; q)_{\infty}}{(c; q)_{\infty}} \frac{(cq^n; q)_{\infty}}{(bq^n; q)_{\infty}}$$

and expand the last quotient by Theorem 3.3:

$$\frac{(cq^n; q)_{\infty}}{(bq^n; q)_{\infty}} = \sum_{k=0}^{\infty} \frac{(c/b; q)_k}{(q; q)_k} (bq^n)^k.$$

Absolute convergence permits interchange of the  $n$ - and  $k$ -sums. The inner  $n$ -sum is another  $q$ -binomial theorem:

$$\sum_{n \geq 0} \frac{(a; q)_n}{(q; q)_n} (zq^k)^n = \frac{(azq^k; q)_{\infty}}{(zq^k; q)_{\infty}} = \frac{(az; q)_{\infty}}{(z; q)_{\infty}} \frac{(z; q)_k}{(az; q)_k}.$$

Substitution gives exactly the right-hand side of (4.1). The identity theorem supplies meromorphic continuation.  $\square$

**Corollary 4.3** ( $q$ -Gauss summation). *If  $|c/(ab)| < 1$ , then*

$${}_2\phi_1 \left( \begin{matrix} a, b \\ c \end{matrix}; q, \frac{c}{ab} \right) = \frac{(c/a, c/b; q)_{\infty}}{(c, c/(ab); q)_{\infty}}. \quad (4.2)$$

*Proof.* Set  $z = c/(ab)$  in (4.1). Since  $az = c/b$ , one upper and one lower parameter in the transformed series cancel. The remaining series is

$$\sum_{k \geq 0} \frac{(c/(ab); q)_k}{(q; q)_k} b^k = \frac{(c/a; q)_\infty}{(b; q)_\infty}$$

by Theorem 3.3. Multiplying by Heine's prefactor yields (4.2).  $\square$

**Corollary 4.4** (First terminating  $q$ -Vandermonde sum). *For  $N \in \mathbb{N}_0$ ,*

$${}_2\phi_1 \left( \begin{matrix} q^{-N}, b \\ c \end{matrix}; q, \frac{cq^N}{b} \right) = \frac{(c/b; q)_N}{(c; q)_N}. \quad (4.3)$$

*Proof.* Set  $a = q^{-N}$  in (4.2). The series terminates, so no convergence restriction on its argument remains. On the product side,

$$\frac{(cq^N; q)_\infty}{(c; q)_\infty} = \frac{1}{(c; q)_N}, \quad \frac{(c/b; q)_\infty}{(cq^N/b; q)_\infty} = (c/b; q)_N.$$

$\square$

**Proposition 4.5** (Second terminating  $q$ -Vandermonde sum). *For  $N \in \mathbb{N}_0$ ,*

$${}_2\phi_1 \left( \begin{matrix} q^{-N}, b \\ c \end{matrix}; q, q \right) = b^N \frac{(c/b; q)_N}{(c; q)_N}. \quad (4.4)$$

*Proof.* Multiply both sides by  $(c; q)_N$  and regard them as polynomials in  $b$  of degree at most  $N$ . It is enough to verify the equality at the  $N + 1$  points  $b = cq^j$ ,  $0 \leq j \leq N$ . At such a point, split  $(cq^j; q)_k / (c; q)_k = (cq^k; q)_j / (c; q)_j$  using (2.1); then use the finite  $q$ -binomial theorem on the resulting sum. Explicitly,

$$\begin{aligned} (c; q)_N S_N(cq^j) &= \sum_{k=0}^N \frac{(q^{-N}; q)_k}{(q; q)_k} q^k (cq^k; q)_j (cq^k; q)_{N-k} \\ &= (c; q)_j \sum_{k=0}^N \frac{(q^{-N}; q)_k}{(q; q)_k} q^k (cq^{j+k}; q)_{N-j-k}, \end{aligned}$$

where terms with a negative final length are interpreted as zero after the finite cancellation that produced the line. Reversing the finite product and applying (3.1) gives

$$(cq^j)^N (q^{-j}; q)_N,$$

which equals  $(cq^j)^N (q^{-j}; q)_N = (cq^j)^N (c/(cq^j); q)_N$  and therefore is the claimed right-hand side at  $b = cq^j$ . Polynomial interpolation completes the proof. (The displayed manipulation is also the usual polynomial proof of the second  $q$ -Vandermonde formula.)  $\square$

## 5 A finite Rogers product lemma and $q$ -Pfaff–Saalschütz

**Lemma 5.1** (Rogers finite product lemma). *For  $n \in \mathbb{N}_0$ ,*

$$(at; q)_n (bt; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a; q)_k (b; q)_k t^k (t; q)_{n-k} (abtq^k; q)_{n-k}. \quad (5.1)$$

*Proof.* Put

$$S_{n,k} = \begin{bmatrix} n \\ k \end{bmatrix}_q (a; q)_k (b; q)_k t^k (t; q)_{n-k} (abtq^k; q)_{n-k}, \quad R_n = \sum_{k=0}^n S_{n,k}.$$

We prove by induction that  $R_n = (at; q)_n (bt; q)_n$ . The case  $n = 0$  is immediate. Set  $Q = q^n$  and, in the  $k$ th summand,  $u = q^k$ . Pascal's recurrence (2.4), followed only by the splitting rule (2.1), gives

$$R_{n+1} = (1 - abtQ) \sum_{k=0}^n S_{n,k} E(u), \quad E(u) = 1 - \frac{tQ}{u} + \frac{tQ}{u} \frac{(1 - au)(1 - bu)}{1 - abtu}. \quad (5.2)$$

The required multiplier is  $M = (1 - atQ)(1 - btQ)$ . The following rational certificate is obtained by ordinary expansion and factorization:

$$\mathcal{C}(u) = \frac{Qabt(u-1)(Qt-u)}{u}.$$

It satisfies the exact rational identity

$$\mathcal{C}(qu) \frac{S_{n,k+1}}{S_{n,k}} - \mathcal{C}(u) = (1 - abtQ)E(u) - M, \quad (5.3)$$

where

$$\frac{S_{n,k+1}}{S_{n,k}} = \frac{t(1 - Q/u)(1 - au)(1 - bu)}{(1 - qu)(1 - tQ/(qu))(1 - abtu)}.$$

Equation (5.3) is verified by multiplying by the three displayed denominators; the two sides reduce to the same polynomial in  $u$ . Multiplying it by  $S_{n,k}$  and summing from  $k = 0$  to  $n$  telescopes:

$$\sum_{k=0}^n S_{n,k} ((1 - abtQ)E(q^k) - M) = \sum_{k=0}^n (\mathcal{C}(q^{k+1})S_{n,k+1} - \mathcal{C}(q^k)S_{n,k}) = 0.$$

The lower boundary vanishes because  $\mathcal{C}(1) = 0$ ; the upper one vanishes because  $S_{n,n+1} = 0$ , equivalently because the ratio contains  $1 - Q/u$  at  $u = Q$ . Thus (5.2) gives  $R_{n+1} = MR_n$ , and induction yields  $R_n = (at; q)_n (bt; q)_n$ .  $\square$

**Theorem 5.2** ( $q$ -Pfaff–Saalschütz summation). For  $n \in \mathbb{N}_0$ ,

$${}_3\phi_2 \left( \begin{matrix} a, b, q^{-n} \\ c, abq^{1-n}/c \end{matrix}; q, q \right) = \frac{(c/a, c/b; q)_n}{(c, c/(ab); q)_n}. \quad (5.4)$$

*Proof.* Multiply (5.4) by  $(c; q)_n (c/(ab); q)_n$ . In its  $k$ th summand use (2.3) twice, once for  $(q^{-n}; q)_k$  and once for  $(abq^{1-n}/c; q)_k$ . All signs and powers of  $q$  cancel, and the summand becomes

$$\begin{bmatrix} n \\ k \end{bmatrix}_q (a; q)_k (b; q)_k \left( \frac{c}{ab} \right)^k (c/(ab); q)_{n-k} (cq^k; q)_{n-k}.$$

Consequently the denominator-cleared identity is exactly Lemma 5.1 with  $t = c/(ab)$ . Its left-hand side is  $(c/b; q)_n (c/a; q)_n$ , which is the denominator-cleared right-hand side of (5.4).  $\square$

*Remark 5.3.* The preceding proof is useful structurally: Bailey's lemma rests on a wholly finite polynomial identity, not on an appeal to analytic continuation or an unproved summation formula. This was one of the main proof-repair goals in reorganizing the source archive.



## 6 Jacobi's triple product and the quintuple product

**Theorem 6.1** (Jacobi triple product). *For  $|q| < 1$  and  $z \neq 0$ ,*

$$\sum_{n \in \mathbb{Z}} q^{\binom{n}{2}} z^n = (q; q)_\infty (-z; q)_\infty (-q/z; q)_\infty. \quad (6.1)$$

*Equivalently,*

$$j(z; q) := (q, z, q/z; q)_\infty = \sum_{n \in \mathbb{Z}} (-1)^n q^{\binom{n}{2}} z^n. \quad (6.2)$$

*Both bilateral series converge normally on compact annuli in  $\mathbb{C}^\times$ .*

*Proof.* Put

$$F(z) = (-z; q)_\infty (-q/z; q)_\infty.$$

It is analytic on  $\mathbb{C}^\times$  and has a normally convergent Laurent expansion  $F(z) = \sum_{n \in \mathbb{Z}} c_n z^n$ . Directly from the products,

$$F(qz) = \frac{1+z^{-1}}{1+z} F(z) = z^{-1} F(z).$$

Comparing Laurent coefficients gives  $c_{n+1} = q^n c_n$ , hence  $c_n = q^{\binom{n}{2}} c_0$  for every  $n \in \mathbb{Z}$ .

It remains to determine  $c_0$ . Euler's expansion (3.5) gives

$$(-z; q)_\infty = \sum_{r \geq 0} \frac{q^{\binom{r}{2}} z^r}{(q; q)_r}, \quad (-q/z; q)_\infty = \sum_{s \geq 0} \frac{q^{s(s+1)/2} z^{-s}}{(q; q)_s}.$$

Therefore

$$c_0 = \sum_{r=0}^{\infty} \frac{q^{r^2}}{(q; q)_r^2}.$$

Take  $c = q$  in the  $q$ -Gauss sum (4.2) and let  $a, b \rightarrow \infty$  after setting the argument equal to  $q/(ab)$ . Termwise,

$$\frac{(a, b; q)_r}{(q, q; q)_r} \left( \frac{q}{ab} \right)^r \rightarrow \frac{q^{r^2}}{(q; q)_r^2},$$

and the product side tends to  $1/(q; q)_\infty$ . Hence  $c_0 = 1/(q; q)_\infty$ , which proves (6.1). Replacing  $z$  by  $-z$  gives (6.2).  $\square$

**Corollary 6.2** (Euler pentagonal theorem). *For  $|q| < 1$ ,*

$$(q; q)_\infty = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(3n-1)/2}. \quad (6.3)$$

*Proof.* In (6.2), replace the base by  $q^3$  and set  $z = q$ . The product is  $(q, q^2, q^3; q^3)_\infty = (q; q)_\infty$ , and the exponent is  $3\binom{n}{2} + n = n(3n-1)/2$ .  $\square$

**Theorem 6.3** (Jacobi quintuple product). *For  $|q| < 1$  and  $z \neq 0$ ,*

$$\begin{aligned} \sum_{n \in \mathbb{Z}} q^{n(3n-1)/2} z^{3n} (1 - zq^n) \\ = (q, z, q/z; q)_\infty (qz^2, q/z^2; q^2)_\infty. \end{aligned} \quad (6.4)$$

*Proof.* Let the product on the right be  $P(z)$ . In terms of  $j$  from (6.2),

$$P(z) = \frac{j(z; q)j(qz^2; q^2)}{(q^2; q^2)_\infty}.$$

The product gives the functional equation

$$P(qz) = q^{-1}z^{-3}P(z).$$

If  $P(z) = \sum_{m \in \mathbb{Z}} c_m z^m$ , then

$$c_{m+3} = q^{m+1}c_m. \quad (6.5)$$

Thus it is enough to determine  $c_0, c_1, c_2$ .

Using the two triple-product series,

$$j(z; q) = \sum_{r \in \mathbb{Z}} (-1)^r q^{r(r-1)/2} z^r, \quad j(qz^2; q^2) = \sum_{s \in \mathbb{Z}} (-1)^s q^{s^2} z^{2s}.$$

Consequently

$$c_0 = \frac{1}{(q^2; q^2)_\infty} \sum_{s \in \mathbb{Z}} (-1)^s q^{3s^2+s} = 1,$$

$$c_1 = -\frac{1}{(q^2; q^2)_\infty} \sum_{s \in \mathbb{Z}} (-1)^s q^{3s^2-s} = -1.$$

Both evaluations are Euler's pentagonal theorem with base  $q^2$ , with  $s \mapsto -s$  in the first. Finally,

$$c_2 = \frac{1}{(q^2; q^2)_\infty} \sum_{s \in \mathbb{Z}} (-1)^s q^{3s^2-3s+1} = 0,$$

because the terms indexed by  $s$  and  $1-s$  cancel. The recurrence (6.5) now gives

$$c_{3n} = q^{n(3n-1)/2}, \quad c_{3n+1} = -q^{n(3n+1)/2}, \quad c_{3n+2} = 0,$$

for all  $n \in \mathbb{Z}$ . This is exactly the Laurent series on the left of (6.4).  $\square$

## 7 Partitions and coefficient recurrences

**Definition 7.1.** A partition of  $n$  is a weakly decreasing finite sequence of positive integers with sum  $n$ . Write  $p(n)$  for the number of partitions of  $n$ , with  $p(0) = 1$  and  $p(n) = 0$  for  $n < 0$ .

**Theorem 7.2** (Euler's partition generating function). For  $|q| < 1$ ,

$$\sum_{n=0}^{\infty} p(n)q^n = \frac{1}{(q; q)_\infty}. \quad (7.1)$$

*Proof.* For each possible part  $m$ , the geometric series  $1 + q^m + q^{2m} + \cdots$  records its multiplicity. Multiplying for all  $m \geq 1$  gives

$$\prod_{m \geq 1} \frac{1}{1 - q^m}.$$

The coefficient of a fixed  $q^n$  receives contributions from only finitely many choices, so the argument is valid formally; for  $|q| < 1$  it is also absolutely convergent.  $\square$

**Corollary 7.3** (Pentagonal recurrence). *For  $n \geq 1$ ,*

$$p(n) = \sum_{k=1}^{\infty} (-1)^{k-1} \left[ p\left(n - \frac{k(3k-1)}{2}\right) + p\left(n - \frac{k(3k+1)}{2}\right) \right], \quad (7.2)$$

where the sum is finite because  $p(m) = 0$  for  $m < 0$ .

*Proof.* Multiply (7.1) by the pentagonal series (6.3). The product is 1. Equating the coefficient of  $q^n$  for  $n > 0$  and isolating  $p(n)$  gives (7.2).  $\square$

**Proposition 7.4** (Partitions in a rectangle). *The coefficient of  $q^m$  in  $\begin{bmatrix} r+s \\ r \end{bmatrix}_q$  is the number of partitions of  $m$  having at most  $r$  parts, each at most  $s$ .*

*Proof.* Let  $G_{r,s}(q)$  denote the indicated generating polynomial. Separate the partitions according to whether they have fewer than  $r$  positive parts or exactly  $r$  positive parts. In the second class subtract 1 from every part; this contributes a factor  $q^r$  and leaves a partition with at most  $r$  parts, each at most  $s-1$ . Hence

$$G_{r,s} = G_{r-1,s} + q^r G_{r,s-1},$$

with  $G_{0,s} = G_{r,0} = 1$ . By (2.5), the Gaussian coefficient  $\begin{bmatrix} r+s \\ r \end{bmatrix}_q$  satisfies the same recurrence and boundary values.  $\square$

*Remark 7.5.* Many finite identities in the archive can be read in two complementary ways: as terminating basic hypergeometric sums, or as identities between weighted partition classes inside rectangles. The analytic route is usually shorter; the finite combinatorial route is often better for positivity.

## 8 Bailey pairs, inversion, and Bailey's lemma

**Definition 8.1** (Bailey pair). Two sequences  $(\alpha_n)_{n \geq 0}$  and  $(\beta_n)_{n \geq 0}$  form a *Bailey pair relative to  $a$*  when

$$\beta_n = \sum_{r=0}^n \frac{\alpha_r}{(q; q)_{n-r} (aq; q)_{n+r}} \quad (n \geq 0). \quad (8.1)$$

**Lemma 8.2** ( $q$ -difference annihilation). *If  $P(x)$  is a polynomial of degree less than  $N$ , then*

$$\sum_{h=0}^N \frac{(-1)^{N-h} q^{\binom{N-h}{2}}}{(q; q)_h (q; q)_{N-h}} P(q^h) = 0. \quad (8.2)$$

*Proof.* By linearity it is enough to take  $P(x) = x^m$ ,  $0 \leq m < N$ . Put  $r = N - h$ . The left-hand side becomes

$$\frac{q^{mN}}{(q; q)_N} \sum_{r=0}^N (-1)^r q^{\binom{r}{2}} \begin{bmatrix} N \\ r \end{bmatrix}_q q^{-mr} = \frac{q^{mN} (q^{-m}; q)_N}{(q; q)_N}$$

by the finite  $q$ -binomial theorem. Since  $0 \leq m < N$ , the product  $(q^{-m}; q)_N$  contains the factor  $1 - q^0$  and is zero.  $\square$

**Theorem 8.3** (Bailey inversion). *The relation (8.1) is equivalent to*

$$\alpha_n = \frac{1 - aq^{2n}}{1 - a} \sum_{j=0}^n \frac{(a; q)_{n+j}}{(q; q)_{n-j}} (-1)^{n-j} q^{\binom{n-j}{2}} \beta_j. \quad (8.3)$$

At  $a = 1$  the formula is interpreted by its removable limit.

*Proof.* Substitute (8.1) into the right-hand side of (8.3) and interchange the two finite sums. The coefficient of  $\alpha_r$  is

$$\frac{1 - aq^{2n}}{1 - a} \sum_{j=r}^n \frac{(a; q)_{n+j} (-1)^{n-j} q^{\binom{n-j}{2}}}{(q; q)_{n-j} (q; q)_{j-r} (aq; q)_{j+r}}. \quad (8.4)$$

If  $r = n$ , the sum has one term and its value is 1. Suppose  $r < n$  and put  $N = n - r$ ,  $h = j - r$ . The quotient of shifted factorials in (8.4) is

$$\frac{(a; q)_{n+j}}{(aq; q)_{j+r}} = (1 - a)(aq^{2r+h+1}; q)_{N-1}.$$

Apart from the harmless factor  $1 - aq^{2n}$ , the coefficient is therefore the left-hand side of (8.2) with

$$P(x) = (aq^{2r+1}x; q)_{N-1},$$

a polynomial of degree  $N - 1$ . It vanishes. Thus the substitution recovers  $\alpha_n$  and no other  $\alpha_r$ , proving that the two triangular relations are inverse.  $\square$

**Corollary 8.4** (The unit Bailey pair). *Relative to  $a$ , the sequence  $\beta_n = \delta_{n,0}$  has inverse*

$$\alpha_n^{\text{unit}}(a) = \frac{1 - aq^{2n}}{1 - a} \frac{(a; q)_n}{(q; q)_n} (-1)^n q^{\binom{n}{2}}. \quad (8.5)$$

For  $a = 1$  this means

$$\alpha_0 = 1, \quad \alpha_n = (-1)^n q^{\binom{n}{2}} (1 + q^n) \quad (n \geq 1). \quad (8.6)$$

*Proof.* Insert  $\beta_0 = 1$  and  $\beta_j = 0$  for  $j > 0$  into (8.3). For  $a \rightarrow 1$ , use

$$\frac{(a; q)_n}{1 - a} \longrightarrow (q; q)_{n-1}, \quad \frac{1 - q^{2n}}{1 - q^n} = 1 + q^n.$$

$\square$

**Theorem 8.5** (Bailey's lemma). *Suppose  $(\alpha_n, \beta_n)$  is a Bailey pair relative to  $a$ . Let  $\rho_1, \rho_2$  be auxiliary parameters and put  $A = aq/(\rho_1\rho_2)$ . Define*

$$\alpha'_n = \frac{(\rho_1, \rho_2; q)_n A^n}{(aq/\rho_1, aq/\rho_2; q)_n} \alpha_n, \quad (8.7)$$

$$\beta'_n = \sum_{j=0}^n \frac{(\rho_1, \rho_2; q)_j (A; q)_{n-j} A^j}{(q; q)_{n-j} (aq/\rho_1, aq/\rho_2; q)_n} \beta_j. \quad (8.8)$$

*Then  $(\alpha'_n, \beta'_n)$  is again a Bailey pair relative to  $a$ .*

*Proof.* Substitute (8.1) into (8.8) and interchange the finite  $j$ - and  $r$ -sums. For a fixed  $r$ , put  $N = n - r$  and  $j = r + s$ . Extract the factors independent of  $s$  using (2.1). Reverse the factor  $(A; q)_{N-s}$  by (2.3). The remaining sum over  $s$  is

$${}_3\phi_2 \left( \begin{matrix} \rho_1 q^r, \rho_2 q^r, q^{-N} \\ aq^{2r+1}, \rho_1 \rho_2 q^{-N}/a \end{matrix}; q, q \right).$$

It is balanced because the second lower parameter is

$$\frac{(\rho_1 q^r)(\rho_2 q^r) q^{1-N}}{aq^{2r+1}} = \frac{\rho_1 \rho_2 q^{-N}}{a}.$$

The  $q$ -Pfaff–Saalschütz sum (5.4) evaluates it. Substituting that product and cancelling the extracted shifted factorials leaves

$$\frac{1}{(q; q)_{n-r}(aq; q)_{n+r}} \frac{(\rho_1, \rho_2; q)_r A^r}{(aq/\rho_1, aq/\rho_2; q)_r} \alpha_r.$$

Summing over  $r$  gives exactly

$$\beta'_n = \sum_{r=0}^n \frac{\alpha'_r}{(q; q)_{n-r}(aq; q)_{n+r}},$$

which is the required Bailey-pair relation.  $\square$

**Corollary 8.6** (Bailey-chain step). *If  $(\alpha, \beta)$  is a Bailey pair relative to  $a$ , then so is*

$$\widehat{\alpha}_n = a^n q^{n^2} \alpha_n, \quad (8.9)$$

$$\widehat{\beta}_n = \sum_{j=0}^n \frac{a^j q^{j^2}}{(q; q)_{n-j}} \beta_j. \quad (8.10)$$

*Proof.* Let  $\rho_1, \rho_2 \rightarrow \infty$  in Bailey's lemma. For fixed  $n$ ,

$$(\rho_1, \rho_2; q)_n \left( \frac{aq}{\rho_1 \rho_2} \right)^n \rightarrow a^n q^{n^2},$$

while all shifted factorials whose arguments tend to zero tend to 1. The formulas reduce to (8.9)–(8.10).  $\square$

**Corollary 8.7** (Weak Bailey lemma). *Under absolute convergence,*

$$\sum_{n=0}^{\infty} a^n q^{n^2} \beta_n = \frac{1}{(aq; q)_{\infty}} \sum_{n=0}^{\infty} a^n q^{n^2} \alpha_n. \quad (8.11)$$

*Proof.* Apply the chain step and write the defining Bailey relation for  $(\widehat{\alpha}, \widehat{\beta})$  at index  $N$ . Let  $N \rightarrow \infty$ . From (8.10),

$$\widehat{\beta}_N \rightarrow \frac{1}{(q; q)_{\infty}} \sum_{j \geq 0} a^j q^{j^2} \beta_j.$$

From the Bailey-pair relation,

$$\widehat{\beta}_N \rightarrow \frac{1}{(q; q)_{\infty}(aq; q)_{\infty}} \sum_{r \geq 0} a^r q^{r^2} \alpha_r.$$

Dominated convergence is justified whenever the two displayed series are absolutely convergent. Multiplication by  $(q; q)_{\infty}$  gives (8.11).  $\square$

**Lemma 8.8** (Parameter lowering from  $a = q$  to  $a = 1$ ). *Suppose  $(\alpha_n, \beta_n)$  is a Bailey pair relative to  $q$ . Define*

$$\widetilde{\alpha}_0 = \alpha_0, \quad (8.12)$$

$$\widetilde{\alpha}_n = (1 - q) \left( \frac{\alpha_n}{1 - q^{2n+1}} - \frac{q^{2n-1} \alpha_{n-1}}{1 - q^{2n-1}} \right), \quad n \geq 1. \quad (8.13)$$

*Then  $(\widetilde{\alpha}_n, \beta_n)$  is a Bailey pair relative to 1.*

*Proof.* Let

$$K_{n,r} = \frac{1}{(q; q)_{n-r}(q; q)_{n+r}}.$$

In  $\sum_{r=0}^n \tilde{\alpha}_r K_{n,r}$ , the coefficient of a fixed  $\alpha_r$  is

$$\frac{1-q}{1-q^{2r+1}} K_{n,r} - \frac{(1-q)q^{2r+1}}{1-q^{2r+1}} K_{n,r+1},$$

where the second term is absent for  $r = n$ . Since

$$\frac{K_{n,r+1}}{K_{n,r}} = \frac{1-q^{n-r}}{1-q^{n+r+1}},$$

the coefficient simplifies to

$$\frac{1-q}{1-q^{n+r+1}} K_{n,r} = \frac{1}{(q; q)_{n-r}(q^2; q)_{n+r}}.$$

This is exactly the kernel of a Bailey pair relative to  $a = q$ . Therefore

$$\sum_{r=0}^n \frac{\tilde{\alpha}_r}{(q; q)_{n-r}(q; q)_{n+r}} = \sum_{r=0}^n \frac{\alpha_r}{(q; q)_{n-r}(q^2; q)_{n+r}} = \beta_n.$$

□

*Remark 8.9.* The elementary lowering step is the missing bridge between the two endpoint Bailey chains  $a = 1$  and  $a = q$ . Placing it at different depths in a chain produces the linear terms in all members of the Andrews–Gordon family.

## 9 The Rogers–Ramanujan identities

**Theorem 9.1** (Rogers–Ramanujan identities). *For  $|q| < 1$ ,*

$$\sum_{n=0}^{\infty} \frac{q^{n^2}}{(q; q)_n} = \frac{1}{(q, q^4; q^5)_{\infty}}, \quad (9.1)$$

$$\sum_{n=0}^{\infty} \frac{q^{n^2+n}}{(q; q)_n} = \frac{1}{(q^2, q^3; q^5)_{\infty}}. \quad (9.2)$$

*Proof.* Start with the unit Bailey pair [Corollary 8.4](#). Apply one Bailey-chain step. Because the original  $\beta$  is  $\delta_{n,0}$ , the new pair has

$$\hat{\beta}_n = \frac{1}{(q; q)_n}, \quad \hat{\alpha}_n = a^n q^{n^2} \alpha_n^{\text{unit}}(a).$$

Apply the weak Bailey lemma to this pair:

$$\sum_{n \geq 0} \frac{a^n q^{n^2}}{(q; q)_n} = \frac{1}{(aq; q)_{\infty}} \sum_{n \geq 0} a^{2n} q^{2n^2} \alpha_n^{\text{unit}}(a). \quad (9.3)$$

First let  $a \rightarrow 1$ . By (8.6), the numerator series on the right is

$$1 + \sum_{n \geq 1} (-1)^n \left( q^{(5n^2-n)/2} + q^{(5n^2+n)/2} \right) = \sum_{m \in \mathbb{Z}} (-1)^m q^{m(5m-1)/2}$$

$$= (q^2, q^3, q^5; q^5)_\infty,$$

where the last equality is Jacobi's triple product with base  $q^5$  and  $z = q^2$ . Division by

$$(q; q)_\infty = (q, q^2, q^3, q^4, q^5; q^5)_\infty$$

gives (9.1).

Now put  $a = q$ . Formula (8.5) becomes

$$\alpha_n^{\text{unit}}(q) = (-1)^n q^{\binom{n}{2}} \frac{1 - q^{2n+1}}{1 - q}.$$

Since  $(q^2; q)_\infty = (q; q)_\infty / (1 - q)$ , the right-hand side of (9.3) is

$$\frac{1}{(q; q)_\infty} \sum_{m \in \mathbb{Z}} (-1)^m q^{m(5m+3)/2}.$$

Indeed, expanding  $1 - q^{2n+1}$  supplies the nonnegative and negative halves of the bilateral sum, with the second term corresponding to  $m = -n - 1$ . The triple product with base  $q^5$  and  $z = q^4$  evaluates the bilateral sum as  $(q, q^4, q^5; q^5)_\infty$ . Cancelling residue classes in  $(q; q)_\infty$  gives (9.2).  $\square$

*Remark 9.2* (Two meanings of the products). The product in (9.1) generates partitions into parts congruent to 1 or 4 modulo 5; the product in (9.2) generates partitions into parts congruent to 2 or 3 modulo 5. The classical difference-two partition interpretations require an additional combinatorial argument. The analytic identities themselves are completely proved above.

## 10 The Rogers–Ramanujan continued fraction

For  $m \geq 0$  define

$$F_m(q) = \sum_{n=0}^{\infty} \frac{q^{n^2+mn}}{(q; q)_n}. \quad (10.1)$$

**Lemma 10.1** (Continuant recurrence). *For  $m \geq 0$  and  $|q| < 1$ ,*

$$F_m = F_{m+1} + q^{m+1} F_{m+2}. \quad (10.2)$$

*Proof.* Subtract the two series  $F_m - F_{m+1}$ . The  $n = 0$  terms cancel, and for  $n \geq 1$  use  $(1 - q^n)/(q; q)_n = 1/(q; q)_{n-1}$ :

$$\begin{aligned} F_m - F_{m+1} &= \sum_{n \geq 1} \frac{q^{n^2+mn}(1 - q^n)}{(q; q)_n} \\ &= \sum_{r \geq 0} \frac{q^{(r+1)^2+m(r+1)}}{(q; q)_r} = q^{m+1} F_{m+2}. \end{aligned}$$

$\square$

**Theorem 10.2** (Rogers–Ramanujan continued fraction). *For  $|q| < 1$ ,*

$$\frac{F_1(q)}{F_0(q)} = \frac{1}{1 + \frac{q}{1 + \frac{q^2}{1 + \frac{q^3}{1 + \ddots}}}}, \quad (10.3)$$

$$R(q) := q^{1/5} \frac{F_1(q)}{F_0(q)} = q^{1/5} \frac{(q, q^4; q^5)_\infty}{(q^2, q^3; q^5)_\infty}. \quad (10.4)$$

*Proof.* Put  $C_m = F_{m+1}/F_m$ . The recurrence (10.2) gives

$$C_m = \frac{1}{1 + q^{m+1}C_{m+1}}. \quad (10.5)$$

Iterating to depth  $M$  gives a finite continuant with tail  $C_M$ . Moreover,  $F_m \rightarrow 1$  as  $m \rightarrow \infty$ : indeed, for  $n \geq 1$ ,

$$\left| \frac{q^{n^2+mn}}{(q; q)_n} \right| \leq \frac{|q|^{n^2+mn}}{(|q|; |q|)_\infty},$$

and the majorant tends summably to zero. Hence  $C_M \rightarrow 1$ . Taking the limit in the finite continuants proves (10.3), including convergence. Finally Theorem 9.1 gives

$$\frac{F_1}{F_0} = \frac{(q, q^4; q^5)_\infty}{(q^2, q^3; q^5)_\infty},$$

which is (10.4). □

## 11 The full Andrews–Gordon family

**Theorem 11.1** (Andrews–Gordon identities). *Let  $k \geq 2$  and  $1 \leq i \leq k$ . For  $n_1, \dots, n_{k-1} \geq 0$  put*

$$N_j = n_j + n_{j+1} + \dots + n_{k-1} \quad (1 \leq j \leq k-1).$$

*Then, for  $|q| < 1$ ,*

$$\begin{aligned} \sum_{n_1, \dots, n_{k-1} \geq 0} \frac{q^{N_1^2 + \dots + N_{k-1}^2 + N_i + \dots + N_{k-1}}}{(q; q)_{n_1} \cdots (q; q)_{n_{k-1}}} \\ = \frac{(q^i, q^{2k+1-i}, q^{2k+1}; q^{2k+1})_\infty}{(q; q)_\infty}. \end{aligned} \quad (11.1)$$

*When  $i = k$ , the linear sum  $N_i + \dots + N_{k-1}$  is empty and equals zero.*

*Proof.* We keep every Bailey operation visible. A chain step applied repeatedly to a pair shows by induction that nested indices are introduced in weakly decreasing order. After  $r$  chain steps and a final weak limit, the  $\beta$ -side has  $r$  indices  $M_1 \geq \dots \geq M_r \geq 0$ , denominator

$$(q; q)_{M_1 - M_2} \cdots (q; q)_{M_{r-1} - M_r} (q; q)_{M_r},$$

and one factor  $a^{M_j} q^{M_j^2}$  for every operation carried out while the relative parameter is  $a$ . This statement follows immediately from (8.10); it is also proved by induction on  $r$ .

*Case  $i = 1$ .* Start with the unit pair relative to  $q$ , perform  $k - 1$  chain steps relative to  $q$ , and apply the weak Bailey lemma relative to  $q$ . All  $k - 1$  nested indices acquire a linear factor  $q^{M_j}$ . With  $n_j = M_j - M_{j+1}$  and  $M_k = 0$ , the  $\beta$ -side is the left-hand side of (11.1) for  $i = 1$ .

On the  $\alpha$ -side, the  $k - 1$  chain steps and the weak limit multiply the unit alpha by  $q^{k(n^2+n)}$ . Since  $(q^2; q)_\infty = (q; q)_\infty / (1 - q)$ , the result is

$$\frac{1}{(q; q)_\infty} \sum_{m \in \mathbb{Z}} (-1)^m q^{((2k+1)m^2 + (2k-1)m)/2}.$$

The negative half is obtained from the term containing  $-q^{2n+1}$  by the substitution  $m = -n - 1$ . Jacobi's triple product, with base  $q^{2k+1}$  and  $z = q^{2k}$ , evaluates the bilateral sum as

$$(q, q^{2k}, q^{2k+1}; q^{2k+1})_\infty,$$



which proves the required product for  $i = 1$ .

Cases  $2 \leq i \leq k$ . Begin again with the unit pair relative to  $q$ . Perform

$$c = k - i + 1$$

chain steps relative to  $q$ , apply the lowering lemma [Lemma 8.8](#), then perform  $i - 2$  chain steps relative to 1, and finally apply the weak Bailey lemma relative to 1. There are  $c + (i - 2) = k - 1$  chain steps in total. The lowering operation does not change  $\beta$ . Exactly the deepest  $c - 1 = k - i$  indices were weighted by the  $q$ -parameter before lowering; after the outer 1-steps and weak limit they are  $M_i, \dots, M_{k-1}$ . Thus the  $\beta$ -side is precisely the left-hand side of [\(11.1\)](#).

It remains to compute the transformed alpha sequence. Before lowering it is

$$\alpha_n = (-1)^n q^{\binom{n}{2} + c(n^2 + n)} \frac{1 - q^{2n+1}}{1 - q}.$$

For  $n \geq 1$ , [\(8.13\)](#) gives

$$\tilde{\alpha}_n = (-1)^n \left( q^{n(n-1)/2 + c(n^2 + n)} + q^{n(n+1)/2 + c(n^2 - n)} \right),$$

while  $\tilde{\alpha}_0 = 1$ . The remaining  $i - 2$  chain steps and the weak limit multiply this by  $q^{(i-1)n^2}$ . Since  $c + i - 1 = k$ , the resulting one-sided sum is the two halves of

$$\sum_{m \in \mathbb{Z}} (-1)^m q^{((2k+1)m^2 + (2k-2i+1)m)/2}. \quad (11.2)$$

Therefore the multiple sum equals [\(11.2\)](#) divided by  $(q; q)_\infty$ . Apply the triple product with base  $q^{2k+1}$  and  $z = q^{2k+1-i}$ . Its product is

$$(q^{2k+1}, q^{2k+1-i}, q^i; q^{2k+1})_\infty,$$

which proves [\(11.1\)](#). □

**Corollary 11.2** (Rogers–Ramanujan as modulus five). *Taking  $k = 2$  and  $i = 2$  or  $i = 1$  in [Theorem 11.1](#) gives [\(9.1\)](#) and [\(9.2\)](#), respectively.*

*Proof.* There is one variable,  $N_1 = n_1$ . For  $i = 2$  the linear term is absent; for  $i = 1$  it is  $n_1$ . The products reduce to the two residue-class products in [Theorem 9.1](#). □

**Corollary 11.3** (The modulus seven family). *For  $r, s \geq 0$ , set  $N_1 = r + s$ ,  $N_2 = s$ . Then the three sums with exponents*

$$N_1^2 + N_2^2 + N_1 + N_2, \quad N_1^2 + N_2^2 + N_2, \quad N_1^2 + N_2^2$$

*are respectively the products obtained by excluding residues  $0, \pm 1$ ,  $0, \pm 2$ , and  $0, \pm 3$  modulo 7.*

*Proof.* This is [Theorem 11.1](#) with  $k = 3$  and  $i = 1, 2, 3$ , after cancelling the three listed residue classes from  $(q; q)_\infty$ . □

**Remark 11.4** (Moduli 9 and 27). The same theorem with  $k = 4$  yields all four odd-modulus Gordon products of modulus 9, and  $k = 13$  yields the thirteen products of modulus 27. This places a large part of the modulus-specific archive under a single proof. Even modulus 14 belongs to the Bressoud-type extension and requires a parity-sensitive Bailey step; it is discussed in [Section 17](#).

## 12 Poisson summation and theta transformations

**Definition 12.1** (Fourier transform). For a Schwartz function  $f: \mathbb{R} \rightarrow \mathbb{C}$ , use the normalization

$$\widehat{f}_{2\pi}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx.$$

**Theorem 12.2** (Poisson summation). For every Schwartz function  $f$ ,

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{m \in \mathbb{Z}} \widehat{f}_{2\pi}(m). \quad (12.1)$$

Both sums converge absolutely.

*Proof.* Periodize  $f$ :

$$P_f(x) = \sum_{n \in \mathbb{Z}} f(x + n).$$

Schwartz decay makes this series, and every termwise derivative, uniformly convergent on compact sets. Thus  $P_f$  is a smooth 1-periodic function. Its  $m$ th Fourier coefficient is, after an absolutely convergent interchange,

$$\int_0^1 P_f(x) e^{-2\pi i m x} dx = \int_{\mathbb{R}} f(x) e^{-2\pi i m x} dx = \widehat{f}_{2\pi}(m).$$

Repeated integration by parts shows that these coefficients decay faster than any power of  $|m|$ , so the Fourier series converges absolutely and uniformly to  $P_f$ . Evaluating it at  $x = 0$  gives (12.1).  $\square$

**Lemma 12.3** (Fourier transform of a Gaussian). For  $t > 0$ ,

$$\int_{\mathbb{R}} e^{-\pi t x^2} e^{-2\pi i x \xi} dx = t^{-1/2} e^{-\pi \xi^2 / t}. \quad (12.2)$$

*Proof.* For  $t = 1$ , let the integral be  $G(\xi)$ . Differentiating under the integral and integrating by parts gives  $G'(\xi) = -2\pi \xi G(\xi)$ . Hence  $G(\xi) = G(0) e^{-\pi \xi^2}$ . The classical polar-coordinate computation

$$G(0)^2 = \iint_{\mathbb{R}^2} e^{-\pi(x^2+y^2)} dx dy = 2\pi \int_0^{\infty} r e^{-\pi r^2} dr = 1$$

shows  $G(0) = 1$ . Scaling  $x \mapsto x/\sqrt{t}$  proves (12.2).  $\square$

For  $\tau$  in the upper half-plane define

$$\begin{aligned} \vartheta_2(\tau) &= \sum_{n \in \mathbb{Z}} e^{\pi i \tau (n+1/2)^2}, \\ \vartheta_3(\tau) &= \sum_{n \in \mathbb{Z}} e^{\pi i \tau n^2}, \\ \vartheta_4(\tau) &= \sum_{n \in \mathbb{Z}} (-1)^n e^{\pi i \tau n^2}. \end{aligned}$$

We take  $(-i\tau)^{1/2}$  to be the holomorphic square root that is positive for  $\tau = it$ ,  $t > 0$ .

**Theorem 12.4** (Theta inversion). For  $\text{Im } \tau > 0$ ,

$$\vartheta_3(-1/\tau) = (-i\tau)^{1/2} \vartheta_3(\tau), \quad (12.3)$$

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2} \vartheta_4(\tau), \quad (12.4)$$

$$\vartheta_4(-1/\tau) = (-i\tau)^{1/2} \vartheta_2(\tau). \quad (12.5)$$

*Proof.* First set  $\tau = it$ ,  $t > 0$ . Apply Poisson summation to  $f(x) = e^{-\pi t x^2}$  and use (12.2):

$$\sum_{n \in \mathbb{Z}} e^{-\pi t n^2} = t^{-1/2} \sum_{m \in \mathbb{Z}} e^{-\pi m^2/t}.$$

This is (12.3) on the positive imaginary axis. Applying Poisson summation to  $f(x + 1/2)$  introduces the Fourier factor  $(-1)^m$  and gives (12.4); applying it to  $(-1)^n f(n)$  is the inverse relation (12.5). Each side is holomorphic in the upper half-plane and the chosen square root is holomorphic there. The identity theorem extends all three equalities from the positive imaginary axis to the whole upper half-plane.  $\square$

**Proposition 12.5** (Theta products). *Put  $Q = e^{\pi i \tau}$ . Then*

$$\vartheta_2(\tau) = 2Q^{1/4} \prod_{n=1}^{\infty} (1 - Q^{2n})(1 + Q^{2n})^2, \quad (12.6)$$

$$\vartheta_3(\tau) = \prod_{n=1}^{\infty} (1 - Q^{2n})(1 + Q^{2n-1})^2, \quad (12.7)$$

$$\vartheta_4(\tau) = \prod_{n=1}^{\infty} (1 - Q^{2n})(1 - Q^{2n-1})^2. \quad (12.8)$$

*Proof.* These are direct specializations of Jacobi's triple product. For example, use base  $Q^2$  in (6.1); setting  $z = 1$  gives (12.7), and setting  $z = -1$  gives (12.8). For  $\vartheta_2$ , factor  $Q^{(n+1/2)^2} = Q^{1/4} Q^{n(n+1)}$  and pair the terms indexed by  $n$  and  $-n-1$  before applying the same product with a shifted argument. The result is (12.6).  $\square$

### 13 The Dedekind eta function and the $q \rightarrow 1^-$ regime

**Definition 13.1** (Dedekind eta). For  $\text{Im } \tau > 0$ ,

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau}).$$

**Lemma 13.2** (Eta and the three theta constants). For  $\text{Im } \tau > 0$ ,

$$2\eta(\tau)^3 = \vartheta_2(\tau)\vartheta_3(\tau)\vartheta_4(\tau). \quad (13.1)$$

*Proof.* Multiply the three product formulas in Proposition 12.5. With  $q = Q^2 = e^{2\pi i \tau}$ , the extra factors besides  $2q^{1/8}(q; q)_{\infty}^3$  cancel because

$$(-q; q)_{\infty}(q; q^2)_{\infty} = \frac{(q^2; q^2)_{\infty}}{(q; q)_{\infty}}(q; q^2)_{\infty} = 1.$$

Since  $\eta(\tau)^3 = q^{1/8}(q; q)_{\infty}^3$ , the result follows.  $\square$

**Theorem 13.3** (Eta transformations). For  $\text{Im } \tau > 0$ ,

$$\eta(\tau + 1) = e^{\pi i / 12} \eta(\tau), \quad (13.2)$$

$$\eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau). \quad (13.3)$$

*Proof.* Equation (13.2) follows directly from the definition, since the infinite product is unchanged and the leading exponential is multiplied by  $e^{\pi i/12}$ .

For inversion, combine (13.1) with the three theta transformations. Their product gives

$$\eta(-1/\tau)^3 = (-i\tau)^{3/2} \eta(\tau)^3.$$

Thus the quotient of the two sides of (13.3) is a holomorphic cube root of unity, hence constant. On the positive imaginary axis all products are positive real and the chosen square root is positive, so the constant is 1.  $\square$

**Theorem 13.4** (Exact modular form of  $(q; q)_\infty$ ). *Let  $t > 0$  and  $q = e^{-t}$ . Then*

$$(q; q)_\infty = \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right) (e^{-4\pi^2/t}; e^{-4\pi^2/t})_\infty. \quad (13.4)$$

Consequently, as  $t \rightarrow 0^+$ ,

$$(q; q)_\infty = \sqrt{\frac{2\pi}{t}} \exp\left(-\frac{\pi^2}{6t} + \frac{t}{24}\right) \left(1 + O(e^{-4\pi^2/t})\right), \quad (13.5)$$

$$\log(q; q)_\infty = -\frac{\pi^2}{6t} + \frac{1}{2} \log \frac{2\pi}{t} + \frac{t}{24} + O(e^{-4\pi^2/t}). \quad (13.6)$$

*Proof.* Put  $\tau = it/(2\pi)$ . From the definition,

$$\eta(\tau) = e^{-t/24} (e^{-t}; e^{-t})_\infty.$$

The eta inversion formula gives

$$\eta\left(\frac{it}{2\pi}\right) = \sqrt{\frac{2\pi}{t}} \eta\left(\frac{2\pi i}{t}\right).$$

Expanding the eta function on the right yields

$$\eta\left(\frac{2\pi i}{t}\right) = e^{-\pi^2/(6t)} (e^{-4\pi^2/t}; e^{-4\pi^2/t})_\infty.$$

Combining these identities proves (13.4). The remaining product differs from 1 by  $O(e^{-4\pi^2/t})$ ; taking a logarithm preserves that error order and gives (13.5)–(13.6).  $\square$

*Remark 13.5.* Formula (13.4), not merely its leading asymptotic term, is a useful numerical acceleration: when  $q$  is close to 1, the transformed nome  $\exp(-4\pi^2/t)$  is exponentially small.

## 14 Ramanujan's bilateral ${}_1\psi_1$ summation

For an integer  $n < 0$ , extend the shifted factorial by

$$(a; q)_n = \frac{1}{(aq^n; q)_{-n}}.$$

This agrees with  $(a; q)_n = (a; q)_\infty / (aq^n; q)_\infty$  whenever both sides are defined.

**Theorem 14.1** (Ramanujan's  ${}_1\psi_1$  sum). *If  $|q| < 1$  and  $|b/a| < |z| < 1$ , then*

$$\sum_{n \in \mathbb{Z}} \frac{(a; q)_n}{(b; q)_n} z^n = \frac{(q, b/a, az, q/(az); q)_\infty}{(b, q/a, z, b/(az); q)_\infty}. \quad (14.1)$$

*Proof.* We first prove the formula for  $b = q^{m+1}$ ,  $m \in \mathbb{N}_0$ . For these values, terms with  $n < -m$  vanish, because the denominator has a pole. Put  $n = k - m$ . Then

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \frac{(a; q)_n}{(q^{m+1}; q)_n} z^n &= z^{-m} \frac{(a; q)_{-m}}{(q^{m+1}; q)_{-m}} \sum_{k=0}^{\infty} \frac{(aq^{-m}; q)_k}{(q; q)_k} z^k \\ &= z^{-m} \frac{(q; q)_m}{(aq^{-m}; q)_m} \frac{(aq^{-m}z; q)_{\infty}}{(z; q)_{\infty}}, \end{aligned}$$

where the last step is the  $q$ -binomial theorem. Repeated use of (2.3) and (2.1) simplifies this expression to

$$\frac{(q, q^{m+1}/a, az, q/(az); q)_{\infty}}{(q^{m+1}, q/a, z, q^{m+1}/(az); q)_{\infty}},$$

which is the right-hand side of (14.1) at  $b = q^{m+1}$ .

Fix  $a, z, q$  with  $0 < |z| < 1$ . On every compact subset of  $|b| < |az|$ , the bilateral series converges normally: the positive tail is controlled by  $|z|^n$ , while after writing  $n = -r$  and using (2.3), the negative tail is controlled by  $|b/(az)|^r$ . Hence both sides of (14.1) are holomorphic functions of  $b$  in that disk, away from removable product singularities. The values  $b = q^{m+1}$  accumulate at 0, so the identity theorem extends the equality to  $|b| < |az|$ , which is exactly  $|b/a| < |z|$ .  $\square$

**Corollary 14.2** (Bilateral-to-product principle). *Any specialization of (14.1) obtained by a limit that is locally uniform on a compact subannulus may be taken termwise. In particular, Jacobi-type bilateral theta sums arise by sending one parameter to 0 or  $\infty$  after a compensating rescaling of  $z$ .*

*Proof.* Normal convergence on compact subannuli permits termwise limits by the Weierstrass theorem. The product side is handled by local uniform convergence of the infinite products.  $\square$

## 15 Finite forms and polynomial approximants

Infinite product identities are often best understood through finite polynomials. Besides giving rigorous convergence, finite forms retain coefficient positivity and are suitable for computer-algebra certification.

**Proposition 15.1** (A finite Rogers–Ramanujan recurrence). *Let  $A_m(q), B_m(q)$  be the numerator and denominator continuants of*

$$\cfrac{1}{1 + \cfrac{q}{1 + \cfrac{q^2}{1 + \cfrac{q^m}{\ddots + \cfrac{q^m}{1}}}}}.$$

*With initial data  $A_{-1} = 1, A_0 = 1$  and  $B_{-1} = 0, B_0 = 1$ , they satisfy*

$$A_m = A_{m-1} + q^m A_{m-2}, \quad B_m = B_{m-1} + q^m B_{m-2}. \quad (15.1)$$

*The finite fraction is  $A_{m-1}/A_m$  after a consistent choice of indexing, and it converges to  $F_1/F_0$  as  $m \rightarrow \infty$ .*

*Proof.* The continuant recurrence follows by clearing the last denominator in a finite continued fraction. Induction proves the stated numerator-denominator recurrences. The tail argument in Theorem 10.2 shows that replacing the true tail  $C_m$  by 1 changes the value by a quantity tending to zero, so the finite continuants converge to  $F_1/F_0$ .  $\square$

**Proposition 15.2** (Coefficientwise passage to infinite products). *Let  $P_N(q)$  be a sequence of polynomials such that, for every fixed degree  $d$ , the coefficients of  $q^0, \dots, q^d$  stabilize as  $N \rightarrow \infty$ . Then  $P_N$  has a well-defined coefficientwise limit in  $\mathbb{C}[[q]]$ . If in addition  $P_N \rightarrow P$  locally uniformly on  $|q| < 1$ , the formal limit is the Taylor series of  $P$ .*

*Proof.* Coefficient stabilization defines a unique formal series. Local uniform convergence allows coefficient extraction on a small circle by Cauchy's integral formula; the coefficient limits must equal the Taylor coefficients of  $P$ .  $\square$

*Remark 15.3.* This elementary principle is the rigorous link between finite Gaussian polynomial identities and their infinite Rogers–Ramanujan or Andrews–Gordon limits. It also prevents a common error in experimental work: pointwise numerical convergence alone does not imply coefficientwise stabilization.

## 16 Cyclotomic dissections and Borwein-type sign questions

Define the finite product

$$P_n(q) = \frac{(q; q)_{3n}}{(q^3; q^3)_n} = \prod_{j=1}^n (1 - q^{3j-2})(1 - q^{3j-1}). \quad (16.1)$$

There are unique polynomials  $A_n, B_n, C_n \in \mathbb{Z}[x]$  such that

$$P_n(q) = A_n(q^3) - qB_n(q^3) - q^2C_n(q^3). \quad (16.2)$$

Uniqueness follows by collecting exponents modulo 3.

**Proposition 16.1** (Reciprocity constraints). *The three residue polynomials satisfy*

$$A_n(x) = x^{n^2} A_n(x^{-1}), \quad (16.3)$$

$$B_n(x) = x^{n^2-1} C_n(x^{-1}), \quad (16.4)$$

$$C_n(x) = x^{n^2-1} B_n(x^{-1}). \quad (16.5)$$

In particular,  $B_n(1) = C_n(1)$  and  $A_n(1) = 2B_n(1)$  for  $n \geq 1$ .

*Proof.* The degree of  $P_n$  is

$$\sum_{j=1}^n [(3j-2) + (3j-1)] = 3n^2.$$

Each pair of factors is reciprocal with positive sign, so  $P_n(q) = q^{3n^2} P_n(q^{-1})$ . Substitute (16.2) on both sides and write  $x = q^3$ . Comparing the three residue classes modulo 3 gives (16.3)–(16.5). Putting  $x = 1$  gives  $B_n(1) = C_n(1)$ . Finally  $P_n(1) = 0$  and (16.2) yield  $A_n(1) = B_n(1) + C_n(1)$ .  $\square$

**Conjecture 16.2** (Borwein nonnegativity). *For every  $n \geq 1$ , all coefficients of  $A_n(x)$ ,  $B_n(x)$ , and  $C_n(x)$  in (16.2) are nonnegative.*

*Remark 16.3.* The conjecture is deliberately stated here without a claim about its complete current status. The archive contains several proved subfamilies, finite checks, and stronger variants with additional parameters. Those variants should not be conflated: a proof for one dissection or one residue class does not automatically settle the full three-polynomial assertion above.

## 17 How the modulus-specific material fits the general machinery

| Archive theme                                  | Structural source                                                         | Treatment in this article                                                                                                                                              |
|------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite $q$ -products and Gaussian coefficients | Finite $q$ -binomial theorem, Cauchy augmentation, polynomial reciprocity | Fully proved in <a href="#">Theorem 3.1</a> , <a href="#">Lemma 3.6</a> , and <a href="#">Proposition 7.4</a> .                                                        |
| Basic hypergeometric summations                | Heine transformation, $q$ -Gauss, $q$ -Vandermonde, $q$ -Pfaff–Saalschütz | Fully proved in <a href="#">Theorems 4.2</a> and <a href="#">5.2</a> , <a href="#">Corollaries 4.3</a> and <a href="#">4.4</a> , and <a href="#">Proposition 4.5</a> . |
| Theta and eta files                            | Jacobi products plus Poisson summation                                    | Fully proved in <a href="#">Theorems 6.1</a> , <a href="#">6.3</a> , <a href="#">12.4</a> and <a href="#">13.3</a> .                                                   |
| Rogers–Ramanujan and continued fractions       | Unit Bailey pair, one chain step, weak limit, continuants                 | Fully proved in <a href="#">Theorems 9.1</a> and <a href="#">10.2</a> .                                                                                                |
| Moduli 7, 9, 27                                | Odd-modulus Andrews–Gordon theorem                                        | All members follow from the proof of <a href="#">Theorem 11.1</a> .                                                                                                    |
| Modulus 14 and parity restrictions             | Even-modulus Bressoud theory and parity-sensitive Bailey transformations  | Located as a distinct extension; not silently identified with the odd-modulus theorem.                                                                                 |
| Bilateral sums                                 | Ramanujan’s ${}_1\psi_1$ and theta specializations                        | Fully proved in <a href="#">Theorem 14.1</a> .                                                                                                                         |
| Borwein-type signs                             | Cyclotomic residue dissections and reciprocal polynomials                 | Structural constraints are proved in <a href="#">Proposition 16.1</a> ; unproved positivity is labelled as <a href="#">Conjecture 16.2</a> .                           |

### 17.1 Why modulus 14 is genuinely different

Odd-modulus Andrews–Gordon products arise from the bilateral exponent  $(2k + 1)m^2 + \lambda m/2$  and an ordinary triple product. Even-modulus identities introduce a parity condition or a factor such as  $(-q; q)_N$  at the finite level. Algebraically, this changes the Bailey kernel; combinatorially, it changes the admissible frequencies of repeated parts. A correct modulus-14 treatment therefore needs a new lemma, not a formal substitution  $2k + 1 \mapsto 14$  in [Theorem 11.1](#).

### 17.2 A reusable proof protocol

For a new identity suggested by the archive, a reliable protocol is:

1. normalize every shifted factorial and every  ${}_r\phi_s$ ;
2. search for a terminating polynomial form before taking an infinite limit;
3. identify the exact Bailey pair and relative parameter;
4. verify the pair by inversion or a finite summation;
5. only then take weak limits and apply a product theorem;

6. separate coefficientwise convergence from numerical pointwise convergence;
7. label sign claims as conjectural unless a coefficient-preserving argument is supplied.

This protocol is more important than any single formula: it turns a collection of experimental identities into a dependency-controlled theory.

## 18 Frontier problems suggested by the unified viewpoint

The following problems are phrased so that proved statements and speculative claims remain sharply separated.

**Problem 18.1** (Finite Andrews–Gordon certificates). Construct, for every  $(k, i)$ , a finite polynomial identity whose coefficientwise limit is (11.1), whose summands are manifestly nonnegative, and whose proof uses only Gaussian recurrences and a bounded number of telescoping certificates. Determine a canonical choice of truncation for which the product side has a direct cylindric-partition or lattice-path interpretation.

**Problem 18.2** (An elementary even-modulus lowering step). Find a parity-sensitive analogue of Lemma 8.8 that turns the odd-modulus Bailey-chain proof into a proof of the even-modulus Bressoud family, with modulus 14 as the first substantial test case. The desired operator should make the parity constraint visible at the level of the Bailey kernel rather than introduce it only after summation.

**Conjecture 18.3** (Sign-stable Borwein tails). *For each fixed offset  $d \geq 0$ , the coefficients at distance  $d$  from either end of the reciprocal polynomials  $A_n, B_n, C_n$  eventually stabilize as  $n \rightarrow \infty$ , and the stabilized coefficients are nonnegative.*

*Remark 18.4.* The reciprocity proved in Proposition 16.1 makes the two ends equivalent. The conjecture is weaker than full coefficientwise nonnegativity, but it is suited to finite-product stabilization and can be tested without extrapolating the central coefficients.

**Problem 18.5** (Total positivity of Bailey kernels). For  $0 < q, a < 1$ , study minors of the lower-triangular matrix

$$M_{n,r}(a, q) = \frac{1}{(q; q)_{n-r}(aq; q)_{n+r}} \quad (0 \leq r \leq n).$$

Determine whether a natural diagonal normalization makes  $M$  totally positive. A positive answer would turn many coefficient-sign statements into consequences of variation-diminishing linear algebra; a negative answer should identify the first obstructing minor and the correct weaker notion.

**Problem 18.6** (Root-of-unity asymptotics). Extend the exact modular acceleration (13.4) to products and Bailey sums as  $q$  approaches a root of unity radially or tangentially. Separate the dominant dilogarithmic exponential, the algebraic prefactor, and the exponentially small resurgence sectors. The finite forms in Proposition 15.2 should serve as uniform error controls.

**Problem 18.7** (Machine-checkable identity certificates). Formalize the dependency chain

$$q\text{-binomial} \Rightarrow q\text{-Pfaff–Saalschütz} \Rightarrow \text{Bailey lemma} \Rightarrow \text{Andrews–Gordon}$$

in a proof assistant. The finite telescoping certificate in Lemma 5.1 is especially suitable: its verification reduces to normalization of a rational function. A useful library should keep formal power series, analytic convergence, and meromorphic parameter continuation as three distinct layers.

**Problem 18.8** (Bilateral Bailey lattices). Combine the parameter-lowering idea with Ramanujan’s  ${}_1\psi_1$  summation to construct bilateral Bailey chains whose limiting products have prescribed residue classes and parity. Identify when a bilateral chain collapses to a unilateral Andrews–Gordon sum and when genuinely new negative-index contributions survive.



## 19 Proof audit and dependency table

| Result                                           | Dependencies used in its proof                                                                      |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Finite $q$ -binomial theorem                     | Pascal recurrence only.                                                                             |
| Infinite $q$ -binomial theorem                   | Coefficient recurrence and product functional equation.                                             |
| Heine transformation                             | Infinite $q$ -binomial theorem plus an absolutely convergent double-sum interchange.                |
| $q$ -Gauss and<br>$q$ -Vandermonde               | Heine transformation, product splitting, and finite reversal.                                       |
| Rogers finite product lemma                      | Pascal recurrence and an explicit rational summand certificate.                                     |
| $q$ -Pfaff–Saalschütz                            | Rogers finite product lemma after denominator clearing.                                             |
| Jacobi triple product                            | Euler expansions, a Laurent functional equation, and $q$ -Gauss for the constant term.              |
| Quintuple product                                | Two triple products, a three-step Laurent recurrence, and Euler’s pentagonal theorem.               |
| Bailey inversion                                 | A finite $q$ -difference annihilation lemma.                                                        |
| Bailey lemma                                     | $q$ -Pfaff–Saalschütz.                                                                              |
| Rogers–Ramanujan                                 | Unit Bailey pair, one chain step, weak Bailey lemma, and Jacobi triple product.                     |
| Andrews–Gordon                                   | Unit pair relative to $q$ , chain steps, parameter lowering, weak Bailey lemma, and triple product. |
| Theta inversion                                  | Poisson summation and the Gaussian Fourier transform.                                               |
| Eta inversion and $(q; q)_\infty$<br>asymptotics | Theta inversion and Jacobi product formulas.                                                        |
| Ramanujan ${}_1\psi_1$                           | $q$ -binomial theorem at a sequence of parameter values and analytic continuation.                  |

## 20 Conclusion

The central lesson of the archive is not merely that many striking products exist. It is that a small number of finite mechanisms repeatedly generate them. The finite  $q$ -binomial theorem supplies the elementary algebra; Heine’s transformation and the finite Rogers lemma supply the decisive summations; Bailey inversion turns a kernel relation into a reusable transformation; and Jacobi’s product converts bilateral quadratic exponents into residue-class products. Poisson summation then connects the same products to modular transformations and sharp asymptotics.

The parameter-lowering lemma makes the full Andrews–Gordon family especially transparent: the position at which one changes from a Bailey pair relative to  $q$  to one relative to 1 is exactly the position at which the linear terms begin in the multisum. This unifies the modulus 5, 7, 9, and 27 material. At the same time, it explains why even modulus and parity-restricted files need genuinely different machinery. Finally, the separation of theorems from conjectural sign patterns provides a safe foundation for further symbolic experimentation and formal verification.

## A Analytic details used repeatedly

**Lemma A.1** (Normal convergence of shifted-factorial series). *Let  $|q| < 1$ . On compact subsets avoiding denominator zeros, every terminating basic hypergeometric series is rational in its parameters, and every nonterminating series used in this article converges normally in the stated open domain.*

*Proof.* Termination is immediate. For  ${}_2\phi_1$ , the ratio of successive terms tends to  $z$ , so compact subsets of  $|z| < 1$  admit a uniform ratio bound below 1. For bilateral  ${}_1\psi_1$ , the positive and negative tails have limiting ratios  $z$  and  $b/(az)$  after reversal; compact subannuli of  $|b/a| < |z| < 1$  again admit uniform bounds. Products converge normally by [Lemma 2.2](#). These estimates also justify every sum interchange and parameter limit made above.  $\square$

**Lemma A.2** (Removable singularity principle). *Suppose a finite rational identity holds for parameters in a nonempty open set and both sides become polynomials after multiplication by the same explicit denominator. Then the denominator-cleared identity holds for all parameter values, and any value at a zero of the denominator is determined by cancellation or a limit.*

*Proof.* The difference of the denominator-cleared sides is a polynomial vanishing on an open set, hence is identically zero. Dividing back gives equality wherever the denominator is nonzero; cancelled values are removable by continuity.  $\square$

## B Notation index

| Symbol                                   | Meaning                                                                                    |
|------------------------------------------|--------------------------------------------------------------------------------------------|
| $(a; q)_n$                               | finite $q$ -shifted factorial $\prod_{j=0}^{n-1} (1 - aq^j)$                               |
| $(a; q)_\infty$                          | convergent infinite shifted factorial for $ q  < 1$                                        |
| $(a_1, \dots, a_r; q)_n$                 | product of shifted factorials with a common base                                           |
| $\begin{bmatrix} n \\ k \end{bmatrix}_q$ | Gaussian or $q$ -binomial coefficient                                                      |
| ${}_r\phi_s$                             | unilateral basic hypergeometric series in the normalization of <a href="#">Theorem 4.2</a> |
| ${}_1\psi_1$                             | bilateral basic hypergeometric series                                                      |
| $j(z; q)$                                | Jacobi product $(q, z, q/z; q)_\infty$                                                     |
| $\vartheta_2, \vartheta_3, \vartheta_4$  | Jacobi theta constants with nome $e^{\pi i \tau}$                                          |
| $\eta(\tau)$                             | Dedekind eta function                                                                      |
| $(\alpha, \beta)$ relative to $a$        | Bailey pair satisfying (8.1)                                                               |
| $N_j$                                    | tail sum $n_j + \dots + n_{k-1}$ in Andrews–Gordon                                         |

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